3.2 Comparison Probing

Despite the regression probes' low error in log-space, their predictions are too noisy to compare close-by values. In Fig. 3b, we use the regression probe's predicted magnitudes to compare the two numbers in the prompt. We observe across all models that this method degrades when the absolute $\log_2$-ratio of the two numbers drops below 1 (i.e., when the two numbers differ by less than a factor of 2). When $|\log_2(a/b)| < 0.2$, the regression probes' predictions are too noisy to meaningfully compare the numbers, resulting in random chance accuracy.

However, the LLM's internal states also seem to include a comparison of the two numbers. We additionally train a binary classifier on the `cross-notation` dataset. Fig. 4 plots accuracy as a function of layer depth. Higher layers consistently exceed 95% accuracy. Fig. 3c shows the accuracy with respect to the log-ratio of two numbers. We observe that, while classifier probes also suffer when the two numbers are close, they generally perform better than regression-based comparison—and much better for 4 of the 7 open-source models.

The internal states encode not only which number is bigger, but how much bigger it is. We trained additional regression probes to predict $\log_2(a/b)$ from the last hidden state of the prompt. These achieved overall predictive accuracy similar to the regression probes of §3.1 (see Fig. 8 in §G). Their accuracy may still degrade when $a \approx b$, but as a approaches b, comparison based on predicted $\log_2(a/b) > 0$ (Fig. 3d) does not degrade quite as quickly as comparison based on predicted $\log_2(a) >$ predicted $\log_2(b)$ (Fig. 3b). Fig. 10 clearly shows that for each model, the log-ratio comparison method is more accurate at all a/b values. Indeed, it behaves similarly to direct binary classification (Fig. 3c)—except for the 4 models whose binary classifiers are much more robust. Presumably those models are internally performing the actual comparison task requested by our prompt.

3.3 Verbalization

Table 1 shows each LLM's verbalized response on `cross-notation` using a one-shot prompt. Across a variety of 7B–8B open-weight models, the best model reaches only 70% accuracy; the worst is at chance (50%). While few-shot prompting improves performance (cf. §H), we emphasize one-shot performance because numeracy *should be* an innate ability of LLMs and such basic questions should require as few examples as possible.

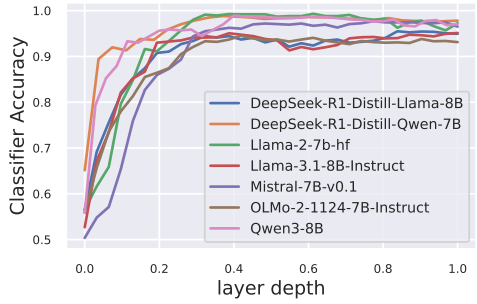

Figure 4: Logistic classifier accuracies on `cross-notation` of each LLM across layers.

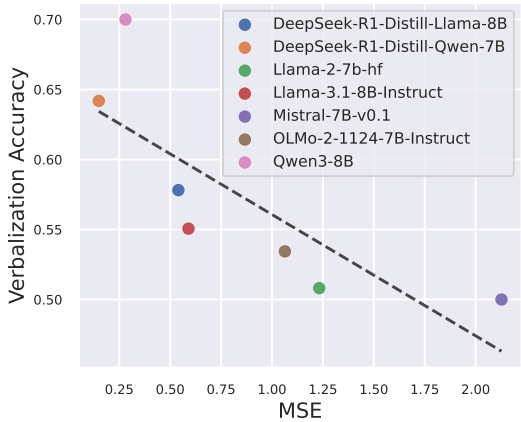

Figure 5: Average performance of linear regression probes at the first 3 layers correlates with model's verbalization accuracy on `cross-notation`.

To visualize the pattern of errors, Fig. 3a shows the verbalized comparison accuracy as $\log_2(a/b)$ changes. Though GPT-4.1 and GPT-4.1-mini achieve over 92% accuracy overall, they have errors at all ratios and especially when the two numbers are close. Interestingly, we observe that Llama-2-7B and Mistral-7B disregard the values of the numbers and almost always answer with the second number. We conduct further experiments in §H and show that, in the one-shot setting, those two models lack awareness of the values of numbers and superficially imitate the example in the one-shot prompt. We present few-shot results and a more thorough error analysis in §H.

3.4 Correlation

To explore how internal representations relate to verbalized performance, we investigate the correlation between probe performance and verbal accuracy across models.

We observe that for scientific notation numerals, the best probes (Figs. 1b–1c) are near the top of

Model	Base	Finetuned	Error Rate Reduction	Finetuned with Probing Loss	Further Error Rate Reduction
DeepSeek-R1-Distill-Llama-8B	57.81	86.69	68.5%	89.69	22.5%
DeepSeek-R1-Distill-Qwen-7B	64.19	84.06	55.5%	95.19	69.8%
Llama-2-7B	50.81	94.44	88.7%	94.56	2.2%
Llama-3.1-8B-Instruct	55.06	96.44	92.1%	95.81	−17.7%
Mistral-7B-v0.1	50.00	95.69	91.4%	99.31	84.0%
OLMo-2-1124-7B	53.44	87.75	73.7%	90.25	20.4%
Qwen3-8B	70.00	96.19	87.3%	99.00	73.8%
Average	57.33	91.61	80.3%	94.83	38.4%

Table 2: Cross-notation comparison accuracy of verbalized responses. Accuracy numbers are percentages without the % sign. Most errors of the base model are fixed by finetuning. Incorporating probing loss into the finetuning objective often fixes many of the *remaining* errors (more than $\frac{2}{3}$ of them for 3 of the models).

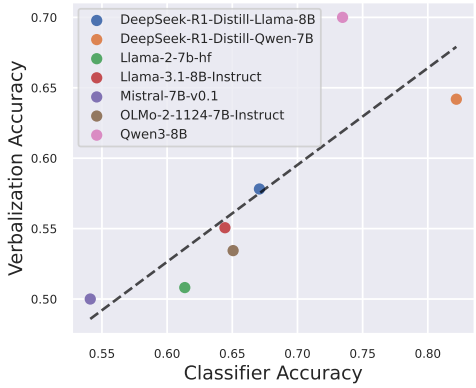

Figure 6: Average performance of logistic classification probes at the first 3 layers predicts the model's verbalization accuracy on `cross-notation`.

the transformer. These representations presumably arrive too late in processing to support the comparisons required by the verbalization task. We further observe that representations that support accurate linear regression are only available after several layers (Fig. 2), and representations that support accurate classification arrive even later (Fig. 4).

We speculate that earlier representations are more useful for downstream tasks. Indeed, Figs. 5 and 6 reveal a strong correlation *across diverse 7B–8B models* between the average performance of probes from the first three layers (even though that performance is still low) and verbalized comparison accuracy on the `cross-notation` dataset.

This correlation suggests that higher quality of internal numerical representations is linked with improved verbalization performance.

3.5 Finetuning

To test whether the link is causal, we incorporate probing loss as an auxiliary objective during finetuning, i.e., we jointly train the LLM and the probe. Table 2 compares three conditions: base model, standard finetuning ($\beta = 0$), and finetuning with probing loss ($\beta = 0.02$).[4] Through hyperparameter tuning, we select the probe layer depth to be 90%. $\beta > 0$ encourages the representations to better support the comparison task that will be needed by verbalization. Through backpropagation, this affects the representations at earlier layers as well.

Ordinary finetuning fixes $> \frac{4}{5}$ of the total errors made by these 7 models (in the sense of reducing error rate). But augmenting the finetuning with probing loss fixes $> \frac{1}{3}$ of the total *remaining* errors. For the 3 Mistral- and Qwen-based models, augmentation fixes $> \frac{2}{3}$ of remaining errors, bringing 2 of them to $\geq 99\%$ accuracy. This suggests the possibility of improving LLMs' verbal numeracy by explicitly improving their internal representations.

4 Conclusion

We study a fundamental question in LLM numeracy (see §A for related work): Do language models know how big numbers are? We found a striking fact: While LLMs do have strong internal numerical representations that can be exposed by both regression and classification probes, they struggle to answer cross-notation comparisons, performing only marginally above chance.

We demonstrate that early-layer representations strongly predict verbalized performance, and auxiliary probing loss enhances finetuning with an additional 3.22% improvement. These results suggest that LLMs possess sophisticated numerical knowledge they cannot reliably access during generation unless fine-tuned for a task, but also that better knowledge causally leads to better generation.

Our work highlights both limitations and opportunities. Current LLMs know more about numbers than they can say; and targeted tuning on these two goals (representation, verbalization) is synergistic.

[4] See hyperparameter tuning in §E.

Limitations

- **Probing reveals representation, not usage.** Unlike causal tracing approaches (Lindsey et al., 2025), we did not investigate how the representations are actually used during inference. Our intervention trained the model to produce both better representations ($\mathcal{L}_{\text{cls}}$) and better text output ($\mathcal{L}_{\text{LM}}$). We found a synergy between these two objectives, but we did not intervene on the representations directly at inference time to see how this changed the text output.

- **Synthetic-real gap.** Most of our core findings—such as probing performance, verbalization failure, and probe-based finetuning gains—are demonstrated on controlled synthetic data. We do not systematically validate these results on real scientific text such as arXiv papers. The extent to which our conclusions generalize to such text (and to other aspects of numeracy) remains to be established.

- **Linear probes may not be enough.** Our tests on arXiv showed that a single linear probe will not necessarily be enough to work beyond our simple synthetic data setting. On real data, we might try using a two-layer neural network, or using the minimum loss of several linear probes that look at different subspaces. Or perhaps these nonlinearities are unnecessary if through finetuning a linear probe, we can successfully change the internal representations so that they do support linear probing after all, even on arXiv. That is, finetuning might recruit a dimension to encode numeric log-magnitude, which might improve the model's abilities in downstream verbal numeracy tasks. We leave these questions to future work.

- **Probing findings are limited to 7B–8B LLMs.** Our probing results are limited to 7B–8B open-weight models where we can access internal representations. We cannot probe GPT-4.1 and GPT-4.1-mini, so it remains unknown whether larger models encode numerical information similarly, or whether the links between internal numeracy and verbal performance are preserved.

- **Limited diversity in test examples.** Our synthetic cross-notation dataset has several constraints. First, all exponents are positive, but negative exponents like 10^{-3} are common in scientific literature. Second, our few-shot demonstration examples (§C) did not include examples where numbers are very close (e.g., 1234 vs. 1.241×10^3); perhaps targeted demonstrations on such cases would reduce their error rate. Third, the same prompt template and few-shot examples are used for all test problems, so our results may depend on those specific ordering and selection patterns. In retrospect, we should have randomly varied the prompt format and few-shot examples within the test set.

- **Semantic context is not considered.** Our probing methods focus on the magnitudes of numerals in isolation, without considering context such as units. In practice, numeracy requires contextual understanding—for instance, 5 miles is greater than 50 meters, even though $5 < 50$, and 50 meters and 50 kg are incomparable. While LLMs likely represent such contextual information internally, we leave this aspect for future work.

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